

Learnings & Reflections Document

1. Team Reflection

The team worked collaboratively with a balanced distribution of responsibilities, allowing each member to contribute meaningfully across both hardware and software tasks. Regular communication and flexibility helped us adapt roles based on immediate needs, which worked well during integration phases. One of our strengths was the ability to support each other across domains rather than working in silos. However, our pace of execution could have been more structured, especially closer to deadlines. We also underestimated the impact of hardware failures. A key component malfunction affected our overall system performance, highlighting the need for backup planning and more rigorous testing at earlier stages.

2. Technical Reflection

One of the main technical challenges was integrating multiple components and code segments into a seamless system. Synchronizing inputs from the ultrasonic sensor with real-time motor control required careful calibration, especially in managing motor speed and responsiveness. On the hardware side, we learned that connections such as jumper wires can be unreliable and need repeated verification. We also experienced component failure due to improper soldering, which emphasized the importance of clean and secure connections. Debugging involved isolating issues step-by-step, testing individual modules, and then reintegrating them, reinforcing a systematic approach to both hardware and software problem-solving.

3. Design Reflection

While the final experience did not fully match our intended interaction due to issues with the ultrasonic sensor, key aspects of the concept were successfully demonstrated. The laser alignment system using mirrors and the LDR worked as intended, and the lighting feedback enhanced the overall experience. The motor-driven motion also functioned reliably in isolation. Users responded positively to the concept, particularly appreciating the aesthetic design and thematic integration. Many expressed interest in experiencing the full interaction as intended. If given more time, we would focus on improving sensor reliability, refining system responsiveness, and ensuring tighter integration between interaction and motion.

4. Personal Reflection

Through this project, I learned how to integrate coding with physical systems, moving beyond working with individual components to building a complete, interactive setup. While we had learned each component separately, combining them into a cohesive system was challenging and required constant testing and iteration. I initially struggled to understand how the electronics would align with the mechanical movement, but over time, both individually and as a team, we were able to bridge that gap.

I also found the model-making process challenging, particularly in achieving precision and stability, but we were able to resolve these issues and create a clean final build. I am most proud of my contribution to the coding and circuitry, and how we collectively worked through problems to bring the project together. With better preparation, especially for hardware uncertainties and code optimization, the final outcome could have been even stronger.